

The empirical recurrence for OEIS A231582, recovered from its tail

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Abstract

OEIS A231582 records “Empirical recurrence of order 72” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 122$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 74$. Its last coefficient is $c_{72} = 11347216957440 \neq 0$, so no further tail satisfies anything shorter; any order-72 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A231582 is “Number of nX3 0..3 arrays with no element less than a strict majority of its horizontal, vertical, diagonal and antidiagonal neighbors.”. It has offset 1 and begins

16, 124, 2352, 32650, 381076, 5618444, 82479418, 1142396851, ...

The entry states:

Empirical recurrence of order 72 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 06:35:18, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 122$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 122$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 72: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 72.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 244$ exact terms from $n = 2$ on, it returns order 72 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{72} c_i a(n-i),$$

c_1	16	c_{19}	−14993049726011	c_{37}	5393236292282338984	c_{55}	−27041529342989406770
c_2	−52	c_{20}	−25502630010801	c_{38}	7489503704892036967	c_{56}	−18091720462012756511
c_3	1040	c_{21}	−49699063274476	c_{39}	7066453536785531103	c_{57}	−7657617330944719739
c_4	−5703	c_{22}	48507657561478	c_{40}	2121200661302550598	c_{58}	−365075394665587672
c_5	−34153	c_{23}	349594617337544	c_{41}	−7738343904127196797	c_{59}	3180947502411003940
c_6	−198733	c_{24}	1145789497037317	c_{42}	−20360468125697176585	c_{60}	3623117316676642116
c_7	−491708	c_{25}	2939801172315949	c_{43}	−30759492445948111997	c_{61}	2629871431196509722
c_8	−534390	c_{26}	6307204386343867	c_{44}	−32417538933950824869	c_{62}	1461446274999843392
c_9	39532610	c_{27}	10489755142452357	c_{45}	−21754239547999813187	c_{63}	604319233263859256
c_{10}	40914366	c_{28}	6251074555124141	c_{46}	−809924060004637621	c_{64}	199723564139620440
c_{11}	664076555	c_{29}	−23721074611693557	c_{47}	23828033300485014527	c_{65}	56316388584534552
c_{12}	4117954450	c_{30}	−110489395356238001	c_{48}	43776275795430722834	c_{66}	13944268265173200
c_{13}	12599922505	c_{31}	−275116797239711345	c_{49}	50149109690094494304	c_{67}	7275888917636256
c_{14}	31861291564	c_{32}	−463311132980443654	c_{50}	41391186231269862380	c_{68}	1906160154937920
c_{15}	6317490954	c_{33}	−532437552711822487	c_{51}	22303990627329494805	c_{69}	−695251068498432
c_{16}	−402862509639	c_{34}	−176678445301378635	c_{52}	−1838074499645646695	c_{70}	−138811039690752
c_{17}	−916563212217	c_{35}	957022273312998556	c_{53}	−20605113365337200859	c_{71}	76841822859264
c_{18}	−3992380098536	c_{36}	2875169711386547269	c_{54}	−28539267443744129521	c_{72}	11347216957440

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 74$, and it is the recurrence OEIS A231582 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{72} - c_1 x^{72-1} - \dots - c_{72}$. Its constant term is $-c_{72} = -11347216957440$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the

components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 72 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 72, so $\deg q = 72$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 72, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 72, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 11347216957440.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-72 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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